

Galaxy-JEPA

Auditing Galaxy Morphology Concepts in a Label-Free Representation

Internal design specification — and paper skeleton

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v2 of Galaxy-Zoo-Classifier (PHYS3004, Nottingham, 2024)

Status — read first. This is a *living internal design document* (Rev 1, skeleton pass). It is organised *as the eventual paper* so it doubles as the publication skeleton, but its present job is to be the single authoritative source of truth the build (and Claude Code) works from — consolidating ~8 scattered scratchpads into one reconciled narrative.

Every non-trivial design point carries a status marker: **[DECIDED]** (locked), **[DEFERRED]** (held until results exist — writing it as prose must not bake in the commitment), or **[OPEN]** (an experimental question to settle before the relevant wiring). Results and Discussion are deliberately *empty scaffolds* — there are no results yet; the point of this pass is to fix the *design* before building. Sections marked **STUB** are structurally placed but their bulk prose is to be consolidated in a later pass.

Contents

1	How to read this document	3
1.1	The status convention	3
1.2	Source documents consolidated here	3
2	Motivation and thesis	3
2.1	The one-paragraph version	3
2.2	Why this, why now — the v1 $\rightarrow$ v2 arc	4
3	The framing decision: A-spine, B-payoff	4
3.1	The two framings	4
3.2	The decision rule (pre-registrable)	5
3.3	Three concrete design consequences	5
4	Core method: labels as a read-out key	5
4.1	The methodological heart	5
4.2	Stage 1 — representation (label-free)	6
4.3	Stage 2 — readout = measurement (the heart)	6
5	The nameability ladder	7
5.1	The diagnosis v1 couldn't give	7
5.2	What sets the rung (besides the feature)	7
6	Uncertainty geometry — the high-beta headline	8
7	The confound taxonomy — Framing B's mechanism	8
7.1	The organising idea	8
7.2	Confound 1 — projection (viewing-angle information loss)	9
7.3	Confound 2 — resolution <i>or</i> semantic ambiguity	9
7.4	Confound 3 — genuine co-occurrence vs artefactual correlation	9
7.5	Why the confused questions are kept, not reduced away	9
8	The feature-scheme experiment	9
8.1	Conditional-population probing — as a comparison, not a fixed mask	9
8.2	The two schemes	10
9	Controls and guardrails	10
9.1	Must-have (Paper 1)	10
9.2	Deferred (Paper 2 / if feasible)	10
10	Statistics — the five decisions, owned	11
11	Evaluation — the deliverables	11
12	The probing harness (locked experimental design)	12
12.1	Two kinds of “locked”	12
12.2	Sub-system 3 — the controls battery	12
12.2.1	3A — the hard gate (the keystone)	12
12.2.2	3B — null-calibrated thresholds + effect-size floor	13
12.2.3	3C — the negative-control battery (five nulls)	13
12.2.4	3D — the nuisance battery (five nuisances + triggered matching)	13
12.3	Sub-system 2 — the ladder	14

12.3.1	2A — entanglement (R1 vs R2): eigen-led triangulation (<i>the strongest machinery</i>)	14
12.3.2	2B/2C — multiplicity and feature scope	14
12.4	Sub-system 4 — uncertainty geometry (the headline)	14
12.5	Sub-system 1 — the full run	15
12.6	Sub-system 5 — the headline figures (deliverables)	15
12.7	New capability this design adds	15
12.8	Implementation state	15
13	Codebase architecture	16
14	Data and multi-survey (Paper 2 scope)	16
14.1	Is “more data → richer representations” even true here?	16
14.2	The homogenisation checklist (worst-first)	16
14.3	The design fork — degrade-down vs preserve-and-correct	17
14.4	The merge as a controlled experiment	17
14.5	Connection to the confound taxonomy	17
15	Prior art and positioning	17
15.1	Galaxy SSL — contrastive, and (newer) masked-reconstruction	17
15.2	Closest neighbour — interpreting galaxy representations against Galaxy Zoo	18
15.3	JEPA — proven adjacent, not (apparently) galaxy morphology	18
15.4	Cross-survey domain gap	18
15.5	Probing / concept-direction lineage (the method’s ancestry)	18
16	Results	18
16.1	Headline figures (planned)	18
16.2	Run order	19
17	Discussion	19
18	Risks and unknowns	19
19	Open questions register	20
20	Parking lot	20

1 How to read this document

This document consolidates the design of Galaxy-JEPA, the second iteration of the undergraduate Galaxy-Zoo-Classifier project. It is both the internal specification the build works from and the skeleton the paper will grow into. The two roles are compatible because the results-independent content of the paper — motivation, method, experimental design, controls, statistics — is exactly the content the build needs settled *now*, while the results-dependent content (which features land where, which framing the data earns) is deliberately deferred.

1.1 The status convention

- **[DECIDED]** — **Decided.** A locked choice. Changing it is a deliberate revision, not a drift. Recorded with its rationale so the reasoning survives.
- **[DEFERRED]** — **Deferred to results.** A choice that *cannot* be made before the data exists (most importantly the paper's spine: method-forward vs taxonomy-audit). Writing these into flowing prose must *not* quietly commit them — the prose states the options and the decision rule, never the outcome.
- **[OPEN]** — **Open experimental question.** A design point that must be settled before the relevant code is wired, but is genuinely undecided now.

1.2 Source documents consolidated here

- `galaxy-jepa-scratchpad.md` (Rev 4) — the *science* source of truth: core idea, nameability ladder, method, uncertainty geometry, controls, multi-survey, paper shape.
- `probing-harness-design.md` — the locked 5-sub-system experimental design of the probing stage.
- `thesis-framing-decision.md` — the A-spine / B-payoff architectural decision.
- `feature-scheme-experiment-design.md` — the feature-scheme experiment and the three-confound interpretive lens (Framing B's mechanism).
- `architecture.md` / `architecture-buildout.md` — the codebase architecture (engineering counterpart).
- the *statistics grounding* (this-session) — the five flagged statistical decisions, now owned.
- `scratchpad-title-and-thesis-note.md` — parked title + thesis-reframe note.

2 Motivation and thesis

2.1 The one-paragraph version

Build a representation of galaxy images **unsupervised** with a JEPA, then use the Galaxy Zoo labels **only as a read-out key** — to name and test directions the representation already learned, never to train it. The labels cannot reshape the encoder's geometry; they can only *misname or blur a read-out direction*. The experiment thus **moves label noise out of representation-learning and into a measurement stage** where it is inspectable and controllable. The scientific question, per feature: is the human morphological concept a *linearly nameable direction*, an *entangled* or *nonlinear* one, or *absent* from the image information at this resolution?

2.2 Why this, why now — the v1 $\rightarrow$ v2 arc

v1’s finding, restated. Galaxy-Zoo-Classifier (the undergraduate dissertation) trained deep CNNs and Vision Transformers on Galaxy Zoo 2 crowdsourced labels, treating the volunteer votes as ground truth. Its central, slightly uncomfortable result: the models performed *to the same ability as the volunteers*, and — crucially — *replicated the volunteers’ confusion*. On the hard questions (bulge shape, winding, arm count) the model confusion matrices mirrored the human ones. The diagnosis: the **labels** are the bottleneck. A supervised model trained to reproduce noisy votes can do no better than the votes themselves; it inherits not just the signal but the noise, including the systematic confusions. Two ViT failure modes made this stark (diss §8): on the confused questions the ViT learned to *always predict zero* (underfitting — it could not extract a feature from the noisy labels), while ResNet50 learned to predict the *average optimal answer* (hedging against confusion). Both are symptoms of the same disease: the labels did not contain clean enough signal to learn from.

The structural problem v2 fixes. Supervised training couples two things that ought to be separate: *learning what a galaxy looks like* and *fitting the (noisy) votes*. Because the loss is computed against the votes, vote-noise flows back through the gradients and *bends the encoder’s weights* — the confusion is baked into the representation. JEPA breaks the coupling: the encoder is trained only on images (predicting masked-region embeddings from context) and never sees a label. Labels enter only afterwards, through a probe that cannot touch the frozen encoder. The noise is thus *relocated* (§4), not removed.

v1’s own reconnaissance feeds v2. The dissertation did not merely note the confusion — it *characterised* it. The §5.2.1 correlation analysis (Figs 18–19) documented, per question, which distinctions volunteers agreed on (edge-on, bar, spiral, odd — strong negative correlations between mutually-exclusive answers) and which they did not (bulge shape, winding, arm count — near-zero off-diagonal structure, i.e. “great confusion”). It even flagged specific cross-feature confusions with physical character: “Disk Edge On” correlating with “Cigar” (Fig 17b), suggesting volunteers could not tell an edge-on disk from a cigar-shaped elliptical. That observation is the seed of this design’s confound taxonomy (§7). v1 saw *that* the features were confused and began to see *why*; v2 has the instrument to measure the why directly.

The arc, in one line. v1 asked “*can we match the volunteers?*”; v2 asks “*can we stop needing them — and use them only to read off what the images already taught the model?*” The motivating pressure is the same one v1 opened with: forthcoming surveys (LSST/Rubin: ~ 150 PB, $\sim 10^{10}$ galaxies) will produce far more imaging than crowdsourcing can ever label. Self-supervised learning on the unlabelled deluge is the field’s answer — and this project asks what such a representation actually *knows* about human morphological concepts, before any label is applied.

3 The framing decision: A-spine, B-payoff

This is the single most important architectural decision about the paper, and it governs how every deferred call below is held.

3.1 The two framings

Framing A — the method. “*Can a label-free representation recover human morphological concepts?*” Contribution is methodological: build a JEPA, freeze it, probe it, produce a per-feature nameability ladder. Subject = the representation. Galaxy Zoo is the *yardstick*. Safe, always-true, promiseable in the abstract regardless of results.

Framing B — the taxonomy audit. “*Which human morphological concepts are real, separable structure in galaxy images — using a label-free representation as the instrument?*” Subject = the

concepts themselves. Galaxy Zoo’s decision tree is the *object under audit*. Bigger, more original — but a much higher defensibility bar, load-bearing on results not yet in hand.

DECIDED

Don’t pick — A is the spine, B is the earned payoff. The paper’s *stated contribution and spine* is Framing A (method + nameability ladder + controls): unimpeachable, true whatever the results. Framing B is the *interpretive arc* — set up early, paid off in the discussion *if and only if the results earn it*. Same discipline as the uncertainty-geometry hedge: backbone on what’s always true; let the bold claims be earned headlines, never load-bearing premises.

3.2 The decision rule (pre-registrable)

DEFERRED — to results

B is earned iff the entanglement on the headline confused features is consistent *across objectives* (JEPA and MAE and MoCo) *and across resolutions* (16×16 and 8×8) — and, added by this design, *across the confound-fingerprint axes* (§7). Consistent $\Rightarrow$ likely the data/taxonomy (B). Inconsistent across objectives $\Rightarrow$ A-only, and the inconsistency is itself a finding (objective-specific entanglement). **Committed at analysis time, not now.**

3.3 Three concrete design consequences

1. **MAE/MoCo baselines: load-bearing for the payoff.** Promoted from “robustness nice-to-have” to “the test that decides whether the paper makes the strong or weak entanglement claim.” They move up the priority order — wanted on the headline confused features.
2. **Entanglement figure (Fig. 3) becomes comparative.** Not “here’s the entanglement” but “here’s whether it is consistent across objectives and resolutions,” because that consistency adjudicates A-vs-B.
3. **Discussion written in two branches, committed at analysis time.** Consistency holds $\rightarrow$ the B payoff; doesn’t $\rightarrow$ honest A-only. Either way the paper is complete and honest.

4 Core method: labels as a read-out key

4.1 The methodological heart

The encoder is built unsupervised and **frozen**. It carves the latent space along whatever axes of variation actually exist in galaxy images. **The labels never bend those axes** — they are consulted only afterwards, to ask “*which pre-existing direction lines up with what humans called a bar / bulge / edge-on disk?*” A mislabelled galaxy can *misname or blur a read-out direction* (a local, inspectable measurement error); it *cannot reshape the encoder’s geometry* the way it deformed v1’s weights. Noise is not eliminated — it is **relocated** from representation-learning to measurement, where it is bounded with controls.

OPEN — experimental question

“Feature = direction” is a **working hypothesis**, not an assumption. The science is *which* features are nameable directions and which are not (the ladder). Held honest by the controls (§9).

4.2 Stage 1 — representation (label-free)

[DECIDED] I-JEPA (Assran et al. 2023): a context-encoder ViT, an EMA target encoder (the anti-collapse mechanism — the target weights are an exponential moving average of the context weights, so they cannot be trivially matched), and a predictor. The model predicts the *embeddings* of masked image blocks from the embeddings of a context block, with the loss an MSE in *latent* space. **There is no pixel reconstruction** — this is the axis that distinguishes I-JEPA from a masked autoencoder (MAE), and it matters for the science: a latent-prediction objective need not preserve pixel-level detail the way a reconstruction objective must, so the two objectives may organise the manifold differently. That difference is exactly what the cross-objective control exploits (§5). Reuse v1’s $256 \times 256 \rightarrow 256 \times (16 \times 16)$ patchification as the tokeniser.

Galaxy-specific design (**[DECIDED]** for the minimal run; later ablations noted):

- **Masking must target the galaxy.** An SDSS cutout is mostly black sky; block-masking it uniformly wastes prediction budget on empty regions. Bias mask sampling toward the galaxy, reusing v1’s average-image bounding box.
- **Rotational symmetry is a free inductive bias.** Galaxies have no canonical orientation. Start with rotation/reflection *augmentation* (simplest, de-risks the minimal run). The principled extension — an **E(2)-equivariant ViT** that bakes in continuous rotation/reflection — reshapes the encoder geometry, so establish the vanilla-ViT ladder first, then ablate. (Precedent: Lens-JEPA, physics-informed, beats supervised + vanilla I-JEPA.)
- **Resolution floor.** The 16×16 token grid ($256 \div 16$) sets the spatial-resolution floor: the model cannot attend below a patch, and the patch-embed linear map compresses sub-patch structure. Thin arms / tight winding may live below that floor — so an **8×8 -patch ablation** is the R4 control (§5), distinguishing *absent from the pixels* from *under-resolved by the tokeniser*.

4.3 Stage 2 — readout = measurement (the heart)

The encoder is **frozen throughout**. Stage 2 is a *measurement protocol*, not a training stage; labels never reach the encoder.

DECIDED

Concept direction — canonically an L2-regularised logistic probe. A feature is linearly nameable if an L2-regularised logistic probe on frozen embeddings reaches statistically significant held-out AUC *and* selectivity over control labels; the unit-normalised probe weight vector is the concept direction. Logistic over mean-difference because it gives a proper decision axis, AUC, regularisation, calibration, and class-imbalance handling. **Keep the mean-difference (CAV) vector as a robustness check** — and note they answer slightly different questions: logistic gives the *discriminative* direction (best separating axis given everything else), mean-difference the *marginal* one. Their *disagreement* is itself an entanglement signal.

Walk the ladder (§5): the linear probe first (defines “clean direction”); then MLP/ k -NN *only* to distinguish entangled-but-present (R3) from absent (R4), and *only* with the controls (§9).

Confidence does double duty — informing the probe and the evaluation, never the encoder — but the two uses must stay *separate axes* to avoid circularity:

- *As a probe target (one axis):* regress the vote fraction, or weight the probe loss by confidence. Useful — but this axis **cannot** be used to test uncertainty geometry (recovering the vote fraction along an axis you trained on it is a tautology).

- *As uncertainty geometry — the headline (a different axis)*: train the concept axis on high-consensus extremes only, then project the held-out ambiguous middle and test whether margin distance ranks vote fractions. The axis never saw the graded values it is asked to reproduce (§6).

[OPEN] Fine-tuning is a subordinate comparison only — unfreezing buys a few points but lets vote-noise bend the encoder again, re-introducing v1’s coupling. A contrast with eyes open, never the main line.

5 The nameability ladder

For any feature, “can the labels name it?” has four answers, each a *different result*:

- R1 Clean linear direction.** A single vector whose projection tracks the feature. (*Bet: smooth/featured, edge-on.*)
- R2 Entangled linear direction.** The axis exists but isn’t orthogonal — moving along “bar” drags bulge-size or inclination. Nameable, not independent.
- R3 Nonlinear manifold.** No single vector; a small MLP/ k -NN decodes it. *Candidate* for “present but tangled” — but see the guardrail.
- R4 Not recoverable.** No probe finds it.

OPEN — experimental question

Guardrail on R3. A nonlinear probe succeeding does *not* by itself mean “present but tangled” — it may have found a *spurious correlate* (survey, redshift, size, brightness, SNR, label prior). R3 is only credible with the controls battery (§9).

5.1 The diagnosis v1 couldn’t give

Mapping the confused features onto the ladder is the paper’s core result:

- Come back as **clean directions (R1)** → supports the strongest version of the v1 thesis: the information was present in the pixels, but the *supervised objective and noisy vote targets* failed to extract it cleanly. (Note: *failed to extract cleanly* — not “purely the labels”; architecture, splits, imbalance, thresholding, augmentation and resolution could also contribute.)
- Need a **(controlled) nonlinear probe (R3)** → present but tangled; supervised training never had a chance through noisy labels.
- **Nothing decodes them (R4)** → genuinely not in the pixels at this resolution; never a labels problem.

Which rung each confused feature lands on is **the paper**.

5.2 What sets the rung (besides the feature)

The rung is **not intrinsic to the feature** — it is a joint property of (*feature* × *SSL objective* × *architecture/resolution* × *probe*). Four confounds, each with its control:

- **Probe capacity / spurious correlates** → the controls battery (§9). A nonlinear probe can win on size / brightness / survey, not morphology.
- **Probe-target circularity** → train the axis on high-consensus *extremes* only, test on the held-out ambiguous middle (§6). Regressing the vote fraction then “recovering” it is a tautology.

- **SSL-objective geometry** → run the *same ladder* on JEPA vs contrastive vs MAE. JEPA’s latent-MSE objective may not flatten semantic manifolds the way contrastive’s push-pull does, so an R3 result in JEPA could be R1-in-contrastive. Cross-objective comparison separates *intrinsic to the images* from *artefact of this objective*. (This is the same machinery that adjudicates Framing B, §3.)
- **Patch size / resolution** → the 8×8 ablation. Thin arms / tight winding living below the token grid can read as R4 (absent) when they are merely under-resolved. This is the R4 control, as the baselines are the R3 control.

Attribution is the whole game: a rung means something about *the images* only once these are held down.

6 Uncertainty geometry — the high-beta headline

Not “does the axis classify bars?” but “does *distance* along the unsupervised axis reproduce the volunteer *vote fraction*?” — a stronger diagnostic than AUC (which can be high on a brittle boundary).

DECIDED

The non-circular protocol. Fit the axis on **high-consensus extremes only** (a binary logistic probe on $v > 0.8$ vs $v < 0.2$). Hold the ambiguous middle ($0.2 < v < 0.8$) out of estimation entirely. Then project those unseen middle galaxies and test (Spearman) whether their margin distance ranks their human vote fraction. An axis trained only on confident examples ordering the ambiguous ones by a human uncertainty it never saw $\Rightarrow$ the geometry genuinely aligns with the human ambiguity gradient, not by construction.

The meaning-flip this enables: v1’s confusion matrices mirrored the volunteers’ *because trained on the votes*; if the latent geometry reproduces that uncertainty *without ever being trained on votes*, the ambiguity is a *real property of the images*, not a labelling artefact. Same observation, opposite conclusion.

DEFERRED — to results

Uncertainty geometry is the highest-upside result *and* the most likely to come out null. The paper is backboned on the ladder + controls + v1 comparison; this is the high-beta headline, not a load-bearing premise.

7 The confound taxonomy — Framing B’s mechanism

DEFERRED — to results

Status of this entire section. This specifies the *mechanism* for Framing B’s earned payoff. It is held as an *interpretive lens pending results*, **not the paper’s spine** (which remains Framing A, §3). The three-confound taxonomy is what B *might deliver if the results earn it* — the same A-spine/B-payoff discipline, one level deeper. It adds a measurement capability (inclination conditioning) but does not revise the locked experimental design (§12). Leading with this story must not read as committing to it.

7.1 The organising idea

A label-free encoder never sees the votes during training, so it is not constrained to reproduce human confusion. This lets us ask, per confused feature: is the confusion in the **data** (the

encoder is also confused $\rightarrow$ a genuine information limit) or in the **humans** (the info is in the pixels; the encoder separates what people cannot)? Grounded in v1’s own correlation analysis, human confusion splits into (at least) three physical causes, each with a distinct fingerprint across the (**inclination** $\times$ **imaging-depth**) plane.

7.2 Confound 1 — projection (viewing-angle information loss)

Case: edge-on disk $\leftrightarrow$ cigar elliptical. v1 data: Edge-on-Disk $\times$ Cigar = +0.83 (strong positive correlation between features on *different* tree branches — the edge-on 2D image cannot disambiguate them). **Fingerprint:** angle-dependent, imaging-depth-*invariant* (no resolution recovers 3D lost to projection). **Test:** condition on inclination.

7.3 Confound 2 — resolution *or* semantic ambiguity

Cases: arm-count, winding, bulge-shape (near-zero off-diagonal structure in v1’s per-question matrices). Two sub-causes the imaging-depth axis *distinguishes*: *resolution* (info in the sky, SDSS can’t resolve it $\rightarrow$ *improves* with deeper imaging) vs *semantic* (the human concept has no sharp boundary $\rightarrow$ imaging depth does nothing). **[OPEN]** Likely per-feature split: arm-count partly semantic (the only non-zero off-diagonal is $3 \leftrightarrow 4$ arms = 0.16); winding more purely resolution; bulge-shape resolution-*or*-subtle. **Test:** recoverability vs imaging depth, per feature.

7.4 Confound 3 — genuine co-occurrence vs artefactual correlation

DEFERRED — to results

Hypothesis — unconfirmed. **Case:** bar + spiral structure (Bar $\times$ 2-arms = +0.56). Consistent with real astrophysical co-occurrence *or* with artefact (clearly-featured galaxies get confident yes-votes downstream). This is the method’s **hard case**, not a settled control: if the encoder shows bar+arms entangled, the entanglement might be *correct* (physics), not a representation limit. Adjudicated by invariance to angle *and* depth plus the astrophysical literature. Cannot be decided until the measurement exists.

7.5 Why the confused questions are kept, not reduced away

v1’s reduction dropped the confused questions out of frustration — but v1 was a *supervised classifier trained on the confused labels*, so it inherited the confusion by construction. v2’s label-free encoder has no such constraint, so the confused questions are **precisely where v2 can beat v1**. Keep the full tree.

8 The feature-scheme experiment

8.1 Conditional-population probing — as a comparison, not a fixed mask

[DECIDED] Each feature is only well-defined within the population that reaches its question (boxy-bulge is meaningless for a no-bulge galaxy — v1’s insight: “Q4 cannot be yes and Q7 cannot be no for the same galaxy”). **But do not hard-mask the “incoherent” galaxies away** — a no-bulge galaxy carrying boxy-bulge votes is a *measurement of human disagreement*, and masking it pre-imposes the tree’s logic before testing whether it holds (circular). Instead, probe each feature across *different* population definitions (full vs consensus-conditional) and **compare**; study the off-population galaxies as their own object (concentrated = systematic confusion = finding; scattered = noise). Reuse the `data/splits.py` firewall machinery.

8.2 The two schemes

- **Scheme 1 — full tree (37 answers, per-bucket)**, each in its conditional population. Honest baseline; expected weak on the confused features (echoes v1 = a finding). BY family ≈ 37 . **Power confound:** per-bucket deep-feature weakness is confounded between genuine-absence and split-sample ($\sim 9,870$ spirals $\div 6$ arm-buckets $\approx 1,600$ each) — Scheme 1 alone cannot distinguish; do not read per-bucket weakness as “absent.”
- **Scheme 2 — reduced/smart (diss-style):** graded questions $\rightarrow$ one graded axis each; binary well-posed $\rightarrow$ one binary feature; odd-subtypes exploratory. BY family $\approx 10\text{--}13$. **Also a power diagnostic:** arm-count-as-axis on the full $\sim 9,870$ recoverable but per-bucket isn’t $\Rightarrow$ the per-bucket failure was power.

[DECIDED] The comparison is a result. Same ladder both ways $\Rightarrow$ reduction cosmetic; differ $\Rightarrow$ reducing changes what’s expressible $\Rightarrow$ a real taxonomy result. **Order: full first** (transparent, no cherry-picking).

OPEN — experimental question

Graded-axis existence test — AUC vs correlation. A graded axis (winding, prominence) may get a *correlation* existence test (projection vs vote-fraction, Spearman/permutation) rather than AUC — but that is the *same measurement* as the uncertainty geometry for that feature, so they may collapse into one thing. Binary features keep AUC regardless. Decide before wiring Scheme 2.

9 Controls and guardrails

These are what separate “we measured something real” from “the probe found a shortcut.” *Attribution is the whole game:* a rung on the ladder (§5) means something about *the images* only once the confounds are held down. The battery is **tiered** to protect a small Paper 1 — must-haves are non-negotiable, the rest defer to Paper 2.

9.1 Must-have (Paper 1)

- **[DECIDED] Selectivity (Hewitt–Liang).** Real-label probe performance *minus* control-label performance. A direction counts only if it beats a control task — this is the guard against probe capacity: a powerful enough probe decodes *anything* (even random labels), so claim strength is tied to the *gap* over a control, not raw performance.
- **[DECIDED] Negative controls.** Concepts that should *not* be clean directions: shuffled vote fractions, random labels, random embeddings, image-metadata bins, sky-background / noise level. The ladder is credible precisely when these *fail* — “bars linear; winding nonlinear; random labels fail; survey identity low after homogenisation” is the shape of a trustworthy result.
- **[DECIDED] Core nuisance probes.** Can the same embedding-axis machinery read off *redshift*, *apparent magnitude*, *angular size* (and survey, if multi-survey)? If a “morphology” direction is really tracking size or brightness, this catches it. This is *the* reason the battery exists: an MLP that “decodes winding” may be decoding apparent size, SNR, or survey, not morphology.

9.2 Deferred (Paper 2 / if feasible)

- **[DEFERRED] Matched evaluation.** Re-test on galaxies matched in redshift / size / brightness, where sample size allows — the strongest form of nuisance control.

- **[DEFERRED] Cross-split robustness.** Learn the direction on one split, test on another; ideally train on SDSS, test on DESI/DECaLS where labels later overlap (needs multi-survey — Paper 2).

The bar. A feature is “linearly nameable” only if the logistic direction clears **AUC + selectivity + the nuisance battery**, not AUC alone. The eigen-triangulation of the entanglement analysis (§12) is the most rigorous machinery in the design and does not depend on the uncertainty-geometry result landing — so the controls plus the ladder plus the v1 comparison form a backbone that is solid whatever the high-beta headlines do.

10 Statistics — the five decisions, owned

The statistical layer is four readings of one picture (a real value vs a chance distribution, “is it past the tail?”) plus a size check. All five flagged decisions are now owned with a defensible rationale.

DECIDED

- (1) **Existence p-value.** Real value (AUC for binary; correlation for graded, pending the open question) vs a null, at the effect floor. **[OPEN]** tie-handling at $AUC \approx 1.0$.
- (2) **Effect floor.** A single pre-registered minimum AUC; the scientific “how strong to call a direction real” call. **[OPEN]** value to be set by -M with rationale.
- (3) **Multiplicity — Benjamini–Yekutieli (FDR).** Chosen knowingly: the existence tests are correlated by construction (bulge levels partition one variable; features co-occur), so independence-assuming methods are wrong; BY controls the false-*discovery* rate (the right target for a discovery catalogue) under arbitrary dependence. Family count per-scheme (≈ 37 full, ≈ 10 –13 reduced).
- (4) **Permutation test** (uncertainty geometry, possibly graded-existence): shuffle vote fractions, recompute Spearman, locate the real value in the shuffle null. $\geq 10,000$ shuffles, two-tailed. **[OPEN]** tie-handling.
- (5) **Marchenko–Pastur null** (entanglement eigenvalues): eigenvalue past the MP edge computed for the actual matrix shape. “Past the edge” to start; Tracy–Widom p-value if a reviewer wants hard significance.

11 Evaluation — the deliverables

A consolidated list of *everything the project measures*. Most items are detailed in their own sections; gathered here so the full deliverable set is visible in one place.

- **[DECIDED] v1 is the baseline.** Same schemes (Hubble / Reduced Set / All Features), same metrics, direct comparability with the existing v1 results table. The v1→v2 continuity is concrete, not rhetorical.
- **[DECIDED] The ladder, per feature** — logistic concept-direction AUC + calibration → MLP/ k -NN selectivity (§5). The core deliverable.
- **[DECIDED] Uncertainty geometry** — projection-vs-vote-fraction (Spearman/Pearson) per feature (§6).
- **[DECIDED] Embedding correlation geometry** — the cosine matrix of concept directions against v1’s Fig 18/19 (§12.3, eigen-triangulation).
- **[DECIDED] Controls** — selectivity, the five negative-control nulls, the five nuisance probes (§9, §12.2). Non-negotiable for Paper 1.

- **[DECIDED] Label-efficiency curve** — probe accuracy vs #labels, SSL-pretrained vs supervised-from-scratch. Quantifies the value of pretraining and directly answers the LSST framing (§4).
- **[DECIDED] SSL baselines = a control, not box-ticking.** Run the *same probe ladder* on JEPA vs MAE (latent prediction vs pixel reconstruction — Wu et al.’s is public) vs **contrastive** (MoCo / SimCLR / BYOL). Cross-objective comparison separates *intrinsic to the images* from *artefact of how JEPA organises the manifold* — and is load-bearing for the Framing-B payoff (§3). A feature that is R3 in JEPA but R1 in contrastive is an objective effect, not an image fact.
- **[DECIDED] Inclination conditioning** — recoverability/entanglement as a function of axis ratio, the confound-fingerprint deliverable (§7).
- **[DEFERRED] Survey-leakage probe** (Paper 2 / multi-survey) — predict source survey from embeddings; drives the merge experiment (§14).
- **[DECIDED] GalaxyMNIST** (Walmsley, 4-class) — external comparability beyond GZ2.

12 The probing harness (locked experimental design)

DECIDED

Hub note. The full derivations live in the authoritative `probing-harness-design.md` (586 lines). This section carries the *actual locked decisions and machinery* at build fidelity — not a summary — because the harness *is* the experimental design (the science), and a reader should not have to open a second document to learn what the controls test or why the eigen-triangulation works. The harness doc remains the source for the full derivations; this section must not contradict it.

12.1 Two kinds of “locked”

[DECIDED] The design distinguishes *structure-locked* (the architecture of a test is fixed) from *formula-flagged* (a specific statistical constant or null-application is -M’s to own before finalising, §10). Everything below is structure-locked; the formula-flagged points are marked.

12.2 Sub-system 3 — the controls battery

12.2.1 3A — the hard gate (the keystone)

[DECIDED] The rung verdict is **not a human reading the numbers** — it is a **pre-registered deterministic function of the controls**, emitted in code, applied identically to every feature, thresholds fixed *before* looking and stamped. Built on the existing **Gate** machinery. The MLP specifically *cannot* return an R3 verdict unless selectivity *and* nuisance-clearance pass — enforced structurally.

Why: controls only protect you if they *gate the conclusion*, not if they merely *accompany* it. If a human assigns the rung after seeing the numbers, “R1” encodes a researcher degree of freedom and the apparatus is decorative. *Cost (accepted):* a feature can land just below the bar and be forced to a rung you “feel” is wrong — that is the point; it is what makes the result honest.

The boundary protocol (sub-threshold-but-feels-present features): when the gate says R4 for a feature intuition says is present, the response is *not* to argue the verdict down. It is that **pre-registered alternative-hypothesis tests fire automatically**, each testing a *named* alternative, each gated on its *own* controls: a controlled MLP (R3 — but only if it clears its own

selectivity + nuisance); the resolution/patch-size ablation (under-resolved, not absent); cross-objective (R4 in JEPA but recoverable in MAE/MoCo = an objective effect). If a pre-registered test recovers it (clearing its controls), the feature moves rung *with a named mechanism*. If none do, it is reported honestly as R4 — and that is a *possible finding*: human-label confidence exceeding image information, the v1-confusion-explained-by-pixels story, the paper’s reason for existing. **The discipline:** “prove it another way” is legitimate *only* when the other way is a pre-registered control testing a named alternative — never “try seventeen probe architectures until one clears the bar.” The rescue-routes are fixed in advance.

12.2.2 3B — null-calibrated thresholds + effect-size floor

[DECIDED]/[OPEN](formula) Thresholds are *not arbitrary constants*. **Existence verdict** = **exceeds the negative-control null at $p < 0.05$** — the bar for “real” is set relative to what the negative controls achieve, adapting to sample size and probe capacity automatically. **Plus a pre-registered effect-size floor** for “clean vs marginal” among real effects (guards the opposite failure: a significant-but-trivial AUC 0.54 at large N). The floor is a pre-registered constant, acceptable *because it no longer does the existence work* — significance-vs-null decides real/not-real, the floor only decides clean-vs-marginal. **Build-order dependency:** the negative-control null is an *input to the gate function* — the gate cannot fire until the null is computed (negatives $\rightarrow$ null $\rightarrow$ real features gated against it).

12.2.3 3C — the negative-control battery (five nulls)

[DECIDED] Existence is calibrated against the **max (most conservative) across all five** — a real feature must beat the *strongest* null. Each closes a distinct attack:

1. **Shuffled vote fractions** — same distribution, destroyed correspondence. Kills “the probe exploits label marginals, not image content.”
2. **Random embeddings** (Gaussian, matched covariance) — kills “any high-D vector predicts this.”
3. **Noise images through the real frozen encoder** (matched to pixel-level statistics, *not* white noise) — kills “the encoder imposes structure on anything.” *The strong one* — “noise images through our encoder give chance AUC” is a clean statement that the encoder learned galaxy structure specifically.
4. **Untrained-encoder embeddings** (frozen random ViT) — kills “the probe, not the pretraining, does the work.” *The headline “pretraining mattered” control* — “a random ViT gets ~ 0.5 on featured-ness, ours gets ~ 0.9 .”
5. **Sky-background / noise-level labels** — kills “the probe reads image depth, not morphology.”

12.2.4 3D — the nuisance battery (five nuisances + triggered matching)

[DECIDED] Answers the most likely *real* confound: morphology genuinely correlates with physical quantities (bars more visible in bigger/closer/brighter galaxies). **The five** (all from the metadata join): redshift, apparent magnitude, Petrosian radius (apparent size), SNR_r (depth), PSF width (seeing). **Adjudication:** a parallel probe predicts each nuisance from the embeddings (diagnostic — shows *whether* there’s a concern); when a nuisance-AUC is comparable-to-or-above the morphology-AUC, that feature *must* be re-tested on galaxies **matched** on that nuisance (held constant, so it can’t be the signal). The feature either *survives* (real) or is *marked confounded*. This makes matching *targeted* (fires only for flagged features — bounded cost), promoting it from

the scratchpad’s “Paper-2/if-feasible” to **Paper-1, targeted**. Every branch is defensible — there is no outcome you’d want to hide.

12.3 Sub-system 2 — the ladder

12.3.1 2A — entanglement (R1 vs R2): eigen-led triangulation (*the strongest machinery*)

[DECIDED] *Not* raw-cosine-vs-arbitrary-threshold (ambiguous between representational entanglement and world-correlation, probe-dependent, trivially passed since random high-D vectors are near-orthogonal). Instead: **several independent measures with different failure modes, triangulated** — where they agree, agreement can’t be an artefact of any single method’s blind spot; where they disagree, the disagreement *localises the cause*.

The eigen spine: effective rank of the concept-direction Gram matrix WW^T (“ k named concepts span only k_{eff} effective dims”; reuses the collapse-monitor code); the **Marchenko–Pastur null** on that spectrum (significantly more entangled than random, by random-matrix theory — a citable null, not a hand-picked threshold); the **eigenvectors** of WW^T (*which* features collapse onto shared axes — localises); the embedding-covariance spectrum (the encoder’s intrinsic dimensionality, as context).

Cross-checks, each covering an eigen blind spot from a different origin: the **cosine matrix** (the human-readable WW^T off-diagonals) — the direct visual bridge to v1, the embedding-recovered matrix laid against v1’s Fig 18/19 human-confusion matrices (“does the representation entangle what humans confused?” — headline Fig 3); **logistic-vs-CAV direction disagreement** (genuinely independent — two different *definitions* of the direction); **conditional recoverability (matched)** — the *causal* attribution, the *only* measure that resolves representation-vs-world (fired surgically, only on eigen-flagged pairs).

DECIDED

Verdict logic (pre-registered — critical). Convergence = strong claim; **disagreements are interpreted, not averaged**, with the interpretation rules fixed *before* results (else triangulation collapses into researcher degrees of freedom). E.g.: eigen-entangled + cosine shows it + logistic-CAV disagree + conditional survives matching → representational entanglement, four methods agree, unassailable. Eigen-entangled *but* conditional vanishes under matching → the entanglement was *world-correlation (astrophysics)*, not representational — itself a clean finding, distinguishable *because* of the causal cross-check. This is the machinery that adjudicates the bar+arms hard case (§7) and Framing B (§3).

[OPEN](formula) The MP-null application and significance machinery on the spectrum are -M’s to own; the structure (triangulation + verdict logic) is locked.

12.3.2 2B/2C — multiplicity and feature scope

[DECIDED] Multiplicity correction is structure-locked, formula per §10 (BY). Feature scope is the two-scheme experiment of §8.

12.4 Sub-system 4 — uncertainty geometry (the headline)

[DECIDED] **Test statistic:** Spearman of projection-distance vs vote-fraction on the held-out ambiguous middle, with a permutation null (§10). **Scope:** gated on R1/R2 (an axis must exist to project along), run on all qualifying features. **4C — the v1 comparison is a first-class deliverable, not an aside:** v1’s supervised-confusion-mirroring laid beside v2’s label-free-geometry-recovery is the clean hook (§6).

12.5 Sub-system 1 — the full run

[DECIDED] **Compute path** is the gate that shapes everything (the medium 40k run vs the rented scale run). **1A:** β stays 0.5 for the headline; the β -sweep is a downstream ablation. **1B:** free probe-replication always; paid encoder-replication only on headline features.

DECIDED

1C — checkpoint choice: label-blind, NOT best-AUC (the subtle one). Do *not* pick the checkpoint that maximises probe AUC — best-by-AUC lets the *labels* (via the probe) choose which encoder you keep, which quietly **breaks the “encoder never saw labels” firewall** (a subtle circularity a sharp reviewer catches). The checkpoint is chosen by a **label-blind** rule, pre-registered before looking at any AUC: the final (most-trained) checkpoint, gated on the collapse trace being stable at the end, else the last-stable checkpoint. This is the one place the intuitive choice (“best performing”) is subtly *wrong*.

12.6 Sub-system 5 — the headline figures (deliverables)

[DECIDED] Specifying the deliverables sub-systems 2–4 *emit*, so figures fall out of the run rather than being reverse-engineered. **Fig 1 — the ladder** (per-feature R1–R4 + AUC + selectivity + calibration + controls-cleared, ordered as narrative). **Fig 2 — uncertainty geometry** (projection vs vote-fraction on the held-out middle, a clean feature against a confused one, with the v1-vs-v2 panel). **Fig 3 — entanglement geometry** (the embedding-recovered cosine matrix against v1’s Fig 18/19, plus the eigenspectrum + MP comparison — the visually striking one). Controls also get figures (generate generously, triage later); the two most likely to earn headline placement are “**untrained encoder ~ 0.5 vs ours ~ 0.9** ” and “**noise-images give chance AUC.**”

12.7 New capability this design adds

Inclination as a conditioning axis (§7) is the one capability the confound work adds; it does *not* revise the locked sub-systems — it is a new axis the existing probe runs along.

- **[DECIDED]** **Inclination proxy = axis ratio (b/a)** — an independent photometric measurement (computed from the pixels by the SDSS pipeline), so conditioning on it to study *vote-confusion* is *not* circular. (Using the T01/T07 votes as the proxy *would* be circular.)
- **[OPEN]**/**metadata gap:** b/a is SDSS photometry (`expAB_r`, `deVAB_r`), *not* a GZ2 vote column, so almost certainly not in the current 40k pull. Fix is a **cheap metadata top-up** — a SQL join on `objID`, few hundred KB, minutes, *no image re-cut*. Pull both `expAB` (disks) and `deVAB` (ellipticals); usage per population is a minor flagged decision.

12.8 Implementation state

[DECIDED] 14 modules scaffolded under `src/galaxy_jepa/probing/` (config, extract, controls, nulls, mlp, entanglement, matching, uncertainty, gates, ladder, run, + `eval/figures.py`), reusing the core firewall (`data/splits.py`), the frozen-encoder Protocol, and the effective-rank callback. The five statistical decisions (§10) sit as parameterised **# FLAGGED** interfaces. 147 tests passing; mypy/ruff clean. The thin vertical slice succeeded (frozen-probe AUC 0.905 on featured-ness), validating the mechanism end-to-end.

13 Codebase architecture

DECIDED

Hub note. The codebase design is the authoritative `architecture.md` / `architecture-buildout.md` pair. Summary here; those remain the source of truth. The governing principle: the *scratchpad* (now this spec, §4–8) is the **science** source of truth; the architecture doc is the **codebase** source of truth. A design choice in the codebase that would change the science is proposed as a science-doc edit, not made silently.

[DECIDED] The architecture borrows the *patterns* of Prism’s M1 (Developer SDK) — its invariants, its normal/power/hard-invariant taxonomy, its gates, and its “the model is just an `nn.Module`” stance — **not** Prism’s platform machinery. (Prism is a separate project; only the design patterns transfer.) The deliberate non-goals matter as much as the goals: this is a single-author research codebase, not a platform — no serving layer, no multi-tenant orchestration, no MLOps machinery beyond a lightweight sweep harness. The heavy declarative-sweep infrastructure earns its keep only at Paper 2’s survey-leakage ablation (§14); building it before the science is de-risked would be premature.

14 Data and multi-survey (Paper 2 scope)

[DEFERRED] Cross-survey transfer is **Paper 2** / **a major ablation, not Paper 1** — kept out of the first clean result, because it is its own deep topic (DeepAstroUDA, AstroCLIP) and because the whole thesis rests on probing being a *trustworthy instrument*.

14.1 Is “more data → richer representations” even true here?

Mostly yes, but *conditionally*. SSL’s premise is that scale and diversity buy generality, and galaxy imaging sits on hundreds of millions of unlabelled cutouts — the regime JEPA wants, one 200k set cannot reach. But “richer” is not automatic: it is gated on the *new* variation being *morphological*, not *instrumental*. Add a second survey and if the dominant new axis the model discovers is “SDSS-ness vs DESI-ness,” capacity has been spent learning the cameras apart. **That gate is the whole game.**

14.2 The homogenisation checklist (worst-first)

Each item is a channel through which *which telescope took the picture* leaks into the representation — and JEPA’s objective will exploit any leak that makes the masked target easier to predict from context (survey identity always does, since context and target share a survey):

- **Pixel scale (arcsec/pixel).** SDSS $\sim 0.396''$, DESI Legacy $\sim 0.262''$, HST $\sim 0.05''$, JWST $\sim 0.03''$. Resizing arrays *looks* like a fix but is not — each crop then covers a different patch of sky. Hold **arcsec/pixel** constant (or kpc/pixel, which needs a redshift), not pixel count.
- **PSF / sharpness.** Ground-based is seeing-limited and blurry; space-based is diffraction-limited and crisp. A space galaxy is intrinsically sharper than any resampled ground-based one — an almost-perfect survey fingerprint.
- **Bandpass + morphological k-correction.** *ugriz* vs *grz* vs near-IR mean different physical things across filter sets; at high z the observed bands sample bluer rest-frame light, so a high- z JWST galaxy is not even the same physical object as a low- z SDSS one in nominally “the same” colours.

- **Depth / SNR / noise texture.** Deeper surveys reveal faint tidal features shallow ones miss; noise statistics differ by instrument.

14.3 The design fork — degrade-down vs preserve-and-correct

Degrade-down: match everyone to the worst common denominator (coarsest PSF, shallowest depth, largest pixel scale). Conservative, kills most leakage, yields a genuinely comparable corpus; cost: throws away the resolution of the good data. **Preserve-and-correct:** keep native quality and make the model invariant to the gap (augmentation spanning it, explicit domain adaptation, or trust scale to wash it out). Higher ceiling, far more to go wrong.

[DECIDED] Recommendation: degrade-down first — you cannot trust a direction that might mean “barred” vs “the survey where bars were easier to see.” Get the comparable corpus and honest diagnostics working; *then*, as a deliberate experiment, relax toward preserve-and-correct and watch leakage and probe quality move.

14.4 The merge as a controlled experiment

The survey-leakage probe turns “did the merge help?” into a measurement, not a hope. Pretrain single-survey vs multi-survey; probe morphology on the *same* held-out labelled set both ways; read **two** numbers off each: **morphology probe AUC** (representation quality) and **survey-leakage AUC** (instrument contamination). *Clean win:* morphology-AUC $\uparrow$ *and* leakage-AUC flat/ $\downarrow$. *The trap (and the interesting failure):* morphology-AUC flat/ $\downarrow$ while leakage-AUC $\uparrow$ — the model spent the extra data on the cameras. Scale to an ablation: plot both AUCs against corpus heterogeneity (single $\rightarrow$ two close surveys $\rightarrow$ add a space-based one).

14.5 Connection to the confound taxonomy

The imaging-depth axis of the confound taxonomy (§7) lives here: **projection-confusion is invariant to imaging depth; resolution-confusion is not.** So the eventual GZ2 $\rightarrow$ DECaLS/DESI comparison is the instrument that *splits* resolution from semantic confusion and *tests* projection’s depth-invariance — a falsifiable, two-direction prediction. **Sequencing:** SDSS + DESI Legacy first (both ground-based optical, overlapping bands, homogenisation tractable); the space-vs-ground jump is experiment two. Don’t reach for JWST to prove the point.

15 Prior art and positioning

OPEN — experimental question

Do not claim firstness. Several IDs/details below are from search snippets and need a proper arXiv sweep before writing. Safest framing: *JEPA-style (latent-prediction) SSL appears underexplored for galaxy morphology relative to contrastive and masked-reconstruction approaches.*

15.1 Galaxy SSL — contrastive, and (newer) masked-reconstruction

- Hayat et al. 2021 — contrastive (MoCo) SSL on SDSS; downstream morphology + photo-z.
- Stein et al. 2022 — MoCo for lens-finding.
- Walmsley et al. 2022 — *Towards Galaxy Foundation Models with Hybrid Contrastive Learning* (BYOL-style + GZ info; introduces **GZ-Evo**, $\sim 552k$ labelled + $\sim 1.34M$ comparable unlabelled). (arXiv 2206.11927)

- Walmsley et al. 2023 — *Galaxy Zoo DESI*, 8.7M galaxies (arXiv 2309.11425); Zoobot.
- AstroCLIP (arXiv 2310.03024) — cross-modal FM, DESI spectra + Legacy images.
- **Masked autoencoders (MAE)** for galaxies (Wu et al., below) — pixel-reconstruction SSL, the natural foil for JEPA’s latent prediction.

15.2 Closest neighbour — interpreting galaxy representations against Galaxy Zoo

Wu et al. 2025 — *Re-envisioning Euclid Galaxy Morphology: Identifying and Interpreting Features with Sparse Autoencoders* (arXiv 2510.23749). SAEs on supervised (Zoobot) and self-supervised (MAE) embeddings, Euclid Q1; finds interpretable features aligned *with and beyond* the GZ decision tree, stronger GZ-alignment than PCA. Public code + MAE. **Our distinctions:**

1. **JEPA latent-prediction vs their MAE pixel-reconstruction** — their MAE is the ideal head-to-head baseline.
2. **Top-down hypothesis-testing** (probe a *named* human concept, walk the ladder) vs **bottom-up discovery** (read features off an SAE dictionary).
3. **Uncertainty geometry** — we test vote-fraction *recovery*, not just label alignment.

Complementary, not competing (one could run an SAE on the JEPA encoder — see §20).

15.3 JEPA — proven adjacent, not (apparently) galaxy morphology

Assran et al. 2023 (I-JEPA); Lens-JEPA (physics-informed gravitational lensing, NeurIPS ML4PS 2025 — beats supervised + vanilla I-JEPA); HEP-JEPA (collider physics, arXiv 2502.03933).

15.4 Cross-survey domain gap

DeepAstroUDA (Ćiprijanović et al., ML:S&T 2023) — frames cross-survey morphology as domain adaptation; warns models extract dataset-specific, non-robust features. Vision-FM-for-astronomy survey (arXiv 2409.11175) — FM features helped optical galaxy classification, poor on radio; transfer isn’t free.

15.5 Probing / concept-direction lineage (the method’s ancestry)

Alain & Bengio 2016 (linear classifier probes); Kim et al. 2018 (TCAV / Concept Activation Vectors); Hewitt & Liang 2019 (control tasks + selectivity).

16 Results

STUB — consolidate in a later pass

EMPTY SCAFFOLD — no results yet. This section is structurally placed for the paper but cannot be written until the experiments run. The planned headline figures are listed below so the scaffold is concrete.

16.1 Headline figures (planned)

1. Concept-direction AUC + calibration per feature — the ladder, quantified.
2. Projection vs vote-fraction per feature (Spearman) — uncertainty geometry.

3. Cosine-similarity matrix between concept directions — entanglement structure (*comparative* across objectives + resolutions, per §3); compare to v1’s Figs 18–19, recovered from embeddings.
4. Recoverability $\times$ inclination (and eventually $\times$ imaging-depth) — the confound-fingerprint figure (§7).
5. Label-efficiency curve — probe accuracy vs #labels, SSL-pretrained vs supervised-from-scratch.

16.2 Run order

Metadata top-up (axis ratio) $\rightarrow$ wire harness (conditional-population capability + inclination axis + finalised stats) $\rightarrow$ **Scheme 1 full-37 medium run on 40k** (code-correctness + signs-of-life gate; deep features *not* yet interpreted — power-starved at 40k, that’s fine, the run’s job is plumbing) $\rightarrow$ Scheme 2 $\rightarrow$ compare $\rightarrow$ [gated, rented scale] real ladder + uncertainty geometry + inclination conditioning + MAE/MoCo baselines $\rightarrow$ [future] DECaLS/DESI imaging-depth comparison.

17 Discussion

DEFERRED — to results

EMPTY SCAFFOLD — written in two branches, committed at analysis time (§3). If cross-objective + cross-resolution + confound-fingerprint consistency holds $\rightarrow$ the Framing-B payoff (“these features may not be separable structure — here’s the label-free evidence, and here’s *why* humans were confused: projection / resolution / semantic”). If it doesn’t $\rightarrow$ the honest Framing-A-only version (“entanglement is objective-specific, a property of these representations, not necessarily the data”). Either way the paper is complete. **Do not pre-write either branch.**

18 Risks and unknowns

The failure modes the design guards against, gathered in one place. Each is either mitigated by a named control or accepted as a bounded risk.

- **Spurious-correlate probes (R3).** A high-capacity probe can decode a feature via nuisance correlates (z, size, brightness, survey, SNR, PSF), not morphology. *The* reason the controls battery exists (§9); an MLP result never stands alone — the hard gate (§12.2, 3A) structurally forbids an R3 verdict without selectivity + nuisance clearance.
- **Probe-capacity confound.** A powerful enough probe decodes *anything* (even random labels). Claim strength is tied to probe simplicity: linear+selective = strongest; MLP-only = weaker and must be controlled. Guarded by selectivity (the gap over a control) and the random-label / random-embedding nulls.
- **Representation collapse** — JEPA’s standard failure mode. Budget tuning on the EMA schedule + masking ratio; the effective-rank collapse monitor (§12) is the detector, and the label-blind checkpoint rule (1C) is gated on a stable end-of-training trace.
- **Survey shortcut** (*multi-survey, Paper 2*) — instrument identity as the dominant learned axis. Caught by the survey-leakage probe; mitigated by homogenisation + close surveys first (§14).

- **Compute + orchestration.** Pretraining a ViT from scratch $\gg$ v1’s transfer-learning runs. The EMA-schedule / masking-ratio grid is *modest* for Paper 1, so a lightweight sweep harness suffices — **do not build MLOps before the science is de-risked**. The heavy declarative-sweep machinery earns its keep only at Paper 2’s survey-leakage ablation. Within FMX hardware, but plan it.
- **Null uncertainty geometry.** The highest-upside result is the most likely to come out null. **[DEFERRED]** Backbone the paper on the ladder + controls + v1 comparison, not on this landing positive (§3, §6).
- **Scope creep.** Stage 1 + frozen logistic probing + must-have controls + the three headline figures is *the minimum that answers the question*. Everything else (E(2) equivariance, multi-survey, fine-tuning, SAE) is explicitly deferred or parked.

19 Open questions register

A consolidated master list of every **[OPEN]** item, for tracking. (To be kept current as the single register; individual sections above hold the context.) Split into *design* questions (settle before the relevant wiring) and *formula-flagged* statistical items (-M’s to own before finalising, structure already locked).

Design questions:

1. Graded-axis existence test: AUC vs correlation (possible merge with uncertainty geometry). Before wiring Scheme 2.
2. Conditional-population gating: which upstream vote, which threshold, consensus vs weight. Per-run knob; default to set.
3. Vote-count filter threshold (v1: $\text{mean}+2\sigma \approx 21$); re-count usable deep-feature N at threshold.
4. Effect-floor value — -M’s scientific call (clean-vs-marginal; not the existence work).
5. Which axis ratio per population (**expAB** vs **deVAB**).
6. Backbone size (ViT-S/16 to start?); from-scratch vs ImageNet warm-start (lean from-scratch); masking specifics; E(2)-equivariance ablation timing.

Formula-flagged (structure locked, formula to own):

7. **Existence significance machinery** (§12.2, 3B) — the specific null-calibrated test and its multiplicity-corrected (BY) form.
8. **Marchenko–Pastur null application** on the concept-direction spectrum (§12.3, 2A) — the edge computation for the actual matrix shape, and whether to add the Tracy–Widom significance.
9. Tie-handling: existence p-value ($\text{AUC} \approx 1$) and the permutation test (§10).
10. Permutation shuffle count and one- vs two-tailed for the uncertainty geometry (lean $\geq 10,000$, two-tailed).

20 Parking lot

Deferred ideas worth recording, *within this project’s scope*: SAE as a disentanglement instrument on the frozen JEPA embeddings (a less-confounded R2/R3 discriminator than an MLP; bridges to Wu et al.); a mechanistic “what did the encoder learn” angle via Jacobian / polar-decomposition

analysis of the layers (a complement to probing — the operator structure rather than the read-out directions); attention-routing (exploratory, with the “attention is not explanation” caveat); and *multimodal pairing of galaxy images with their SDSS spectra* (AstroCLIP-adjacent — a possible future direction, and a separate project in its own right).

End of Rev 1 (skeleton). Bulk-prose STUB sections to be consolidated in subsequent passes; Results and Discussion to be written when the experiments exist.